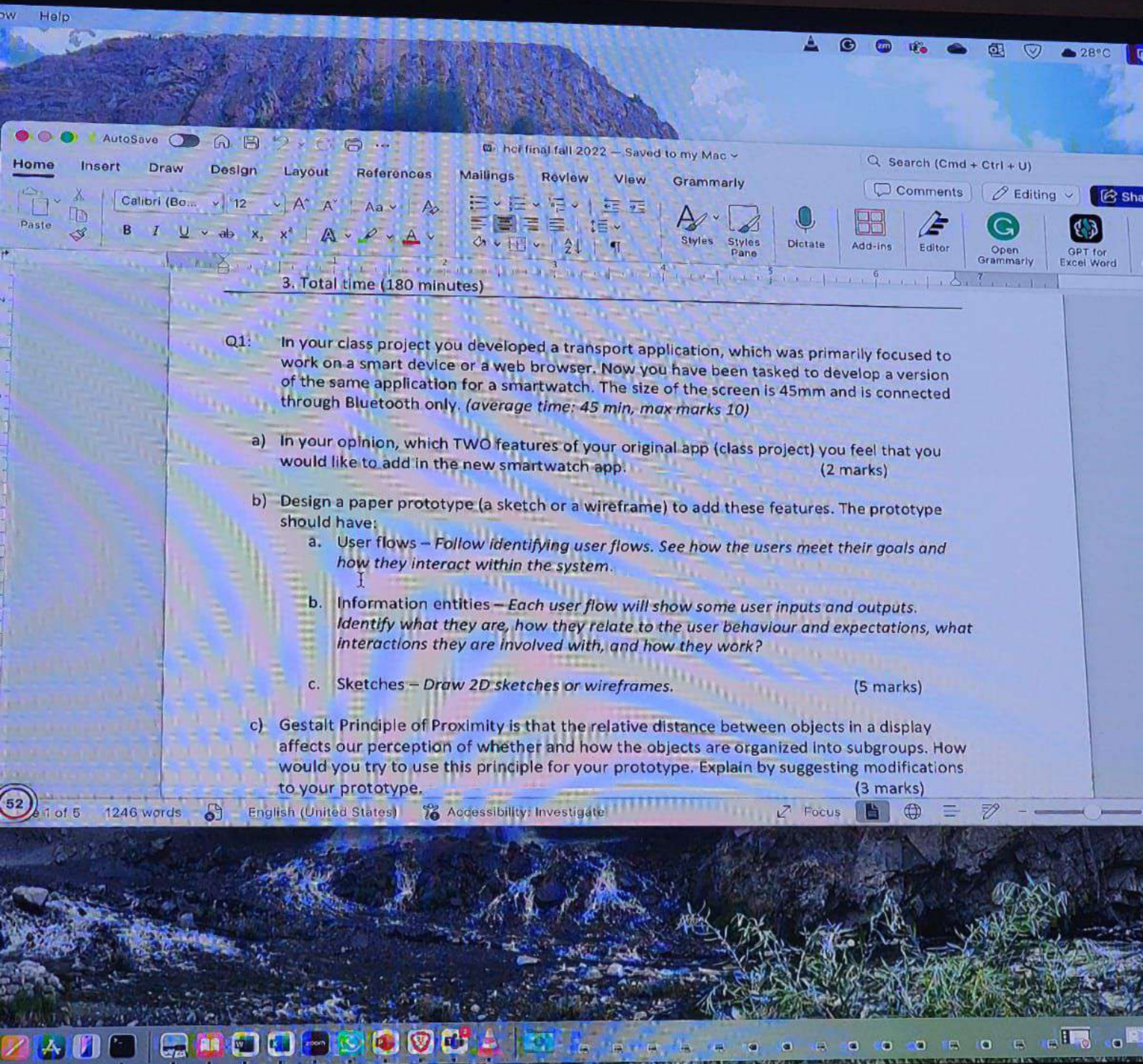
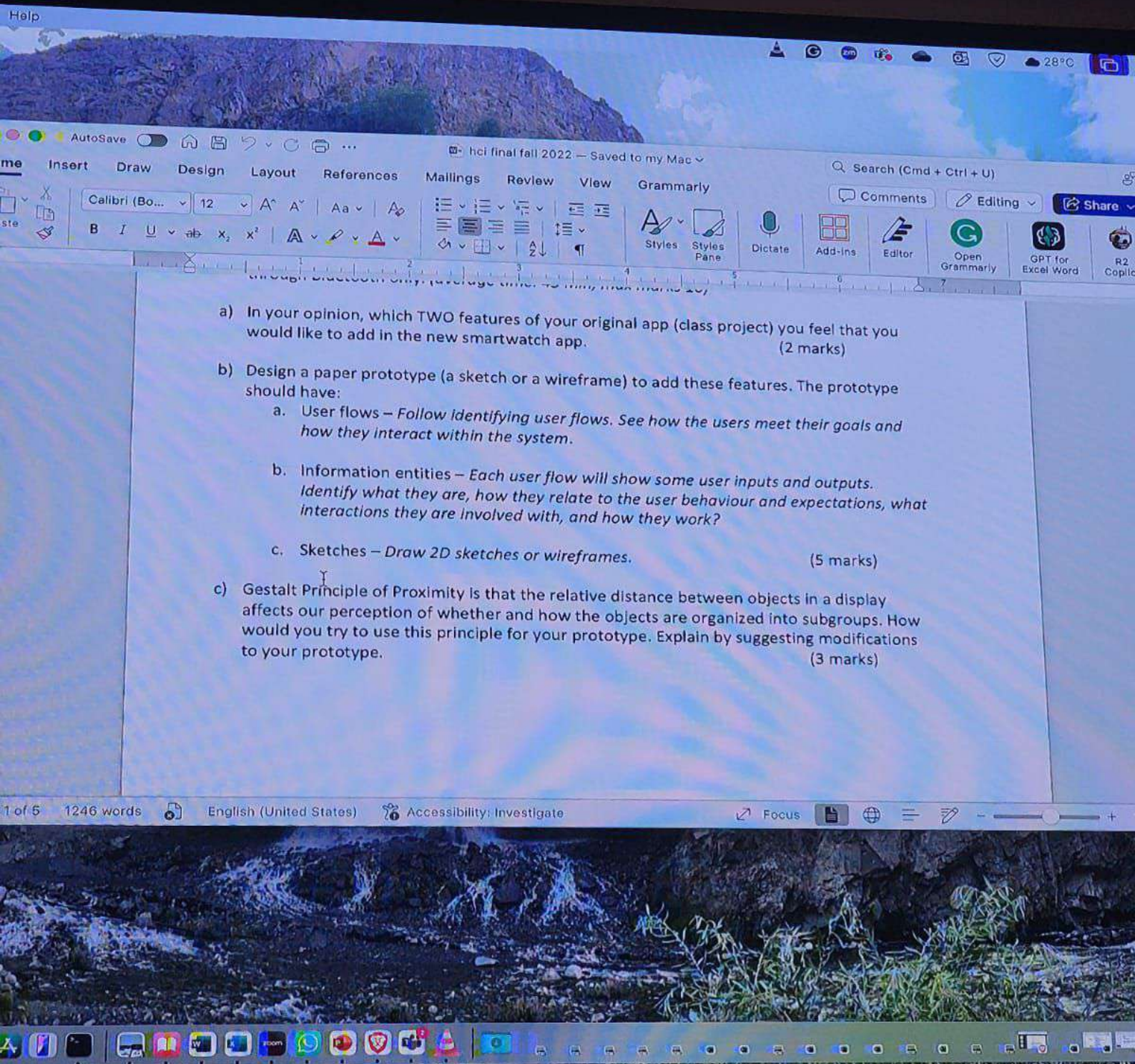


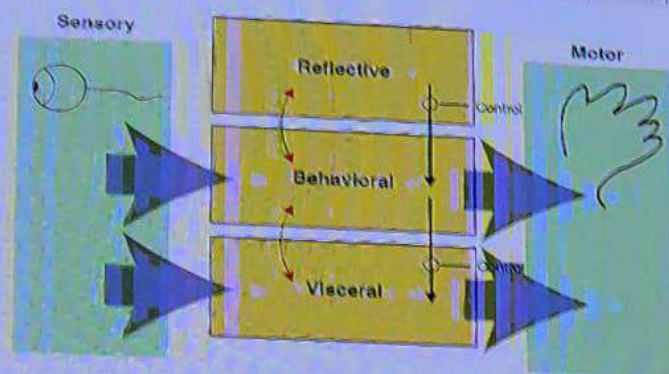


3. Total time (180 minutes)

- Q1: In your class project you developed a transport application, which was primarily focused to work on a smart device or a web browser. Now you have been tasked to develop a version of the same application for a smartwatch. The size of the screen is 45mm and is connected through Bluetooth only. (average time: 45 min, max marks 10)
- In your opinion, which TWO features of your original app (class project) you feel that you would like to add in the new smartwatch app. (2 marks)
 - Design a paper prototype (a sketch or a wireframe) to add these features. The prototype should have:
 - User flows – *Follow identifying user flows. See how the users meet their goals and how they interact within the system.*
 - Information entities – *Each user flow will show some user inputs and outputs. Identify what they are, how they relate to the user behaviour and expectations, what interactions they are involved with, and how they work?*
 - Sketches – *Draw 2D sketches or wireframes.* (5 marks)
 - Gestalt Principle of Proximity is that the relative distance between objects in a display affects our perception of whether and how the objects are organized into subgroups. How would you try to use this principle for your prototype. Explain by suggesting modifications to your prototype. (3 marks)



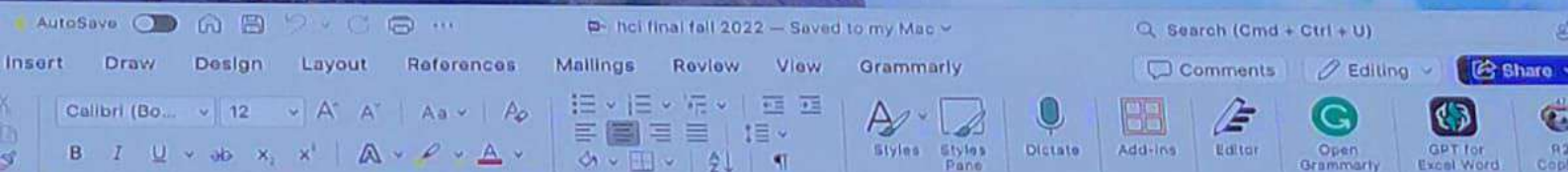
- a) In your opinion, which TWO features of your original app (class project) you feel that you would like to add in the new smartwatch app. (2 marks)
- b) Design a paper prototype (a sketch or a wireframe) to add these features. The prototype should have:
- a. User flows – *Follow Identifying user flows. See how the users meet their goals and how they interact within the system.*
 - b. Information entities – *Each user flow will show some user inputs and outputs. Identify what they are, how they relate to the user behaviour and expectations, what interactions they are involved with, and how they work?*
 - c. Sketches – *Draw 2D sketches or wireframes.* (5 marks)
- c) Gestalt Principle of Proximity is that the relative distance between objects in a display affects our perception of whether and how the objects are organized into subgroups. How would you try to use this principle for your prototype. Explain by suggesting modifications to your prototype. (3 marks)



Imagine, you have asked to influence an e-commerce site (eg. Amazon.com) to increase the sales by 100%.

To influence customers, you need to use some of the following features:

- i. Update the status on social media platforms.
 - ii. Update the goal.
 - iii. Set challenge parameters (date/goal/participation) and ask friends to join a challenge.
 - iv. Set reward parameter.
- a) What changes in your design you will suggest incorporating the combination of emotional model and social media features.
- b) It is commented that recognition is better than recall. Which aspect of your design



d) It is commented that recognition is better than recall, which aspect of your design follows this approach? Explain by giving an example from your project/prototype. (5 marks)

Q2 (b) The development manager of a website for online book-buying section of Amazon.com has asked you to carry out a heuristic evaluation of the usability of the book section. He has specifically proposed the three heuristics listed below.

1. "There should be between five and nine navigation options on each page."
2. "There should be a good match between the navigation buttons and the users' goals."
3. "It should be easy for users to change their plans."

Based on your knowledge of the course, answer the following:

- (a) Has the manager misunderstood heuristic evaluation? Briefly justify your answer.
- (b) Please comment on the above three heuristics suggested by the manager. For each of the proposed heuristics, your comments should include:

- (i) any theoretical justification for (or against) this heuristic;
- (ii) any additional evaluation steps that might be required in applying it; and
- (iii) the likely impact of such evaluation on the system design.

(average time 25 min, max marks 5)

